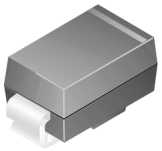
General Purpose Rectifiers (Glass Passivated)

GF1A - GF1M

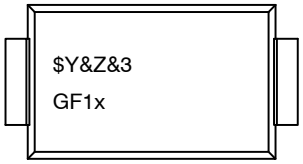


SMA
CASE 403AE
(COLOR BAND DENOTES CATHODE)

Features

- Low Forward Voltage Drop
- High Current Capability
- Easy Pick and Place
- High Surge Current Capability
- This Device is Pb-Free and is RoHS Compliant

MARKING DIAGRAM



$Y	= **onsemi** Logo
&Z	= Assembly Plant Code
&3	= Data Code (Year & Week)
GF1x	= Specific Device Code
x	= A/B/D/G/J/K/M

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

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ABSOLUTE MAXIMUM RATINGS T_A = 25°C unless otherwise noted

Symbol	Parameter	Value							Units
		1A	1B	1D	1G	1J	1K	1M	
V_{RRM}	Maximum Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current, @ T_L = 125°C	1.0							A
I_{FSM}	Non−repetitive Peak Forward Surge Current 8.3 ms Single Half−Sine−Wave	30							A
T_{stg}	Storage Temperature Range	−65 to +175							°C
T_J	Operating Junction Temperature Range	−65 to +175							°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS

Symbol	Parameter	Value	Units
P_D	Power Dissipation	1.8	W
$R_{\theta JA}$	Thermal Resistance, Junction−to−Ambient*	80	°C/W
$R_{\theta JL}$	Thermal Resistance, Junction−to−Lead*	26	°C/W

*Device mounted on PCB with 0.2 x 0.2″ (5.0 x 5.0 mm) copper pad areas.

ELECTRICAL CHARACTERISTICS T_A = 25°C unless otherwise noted

Symbol	Parameter	Device							Units
		1A	1B	1D	1G	1J	1K	1M	
V_F	Maximum Forward Voltage @ 1.0 A	1.0					1.2		V
t_{rr}	Maximum Reverse Recovery Time I_F = 0.5 A, I_R = 1.0 A, I_{rr} = 0.25 A	2.0							μs
I_R	Maximum Reverse Current @ Rated V_R T_A = 25°C T_A = 125°C	5.0 50							μA
C_T	Typical Capacitance V_R = 4.0 V, f = 1.0 MHz	15							pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

PACKAGE MARKING AND ORDERING INFORMATION

Device	Device Marking	Package	Quantity†
GF1A	GF1A	SMA	7500 / Tape & Reel
GF1B	GF1B		
GF1D	GF1D		
GF1G	GF1G		
GF1J	GF1J		
GF1K	GF1K		
GF1M	GF1M		

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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TYPICAL CHARACTERISTICS

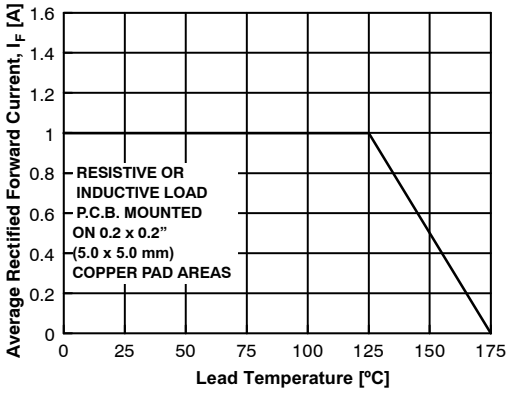


Figure 1. Forward Current Derating Curve

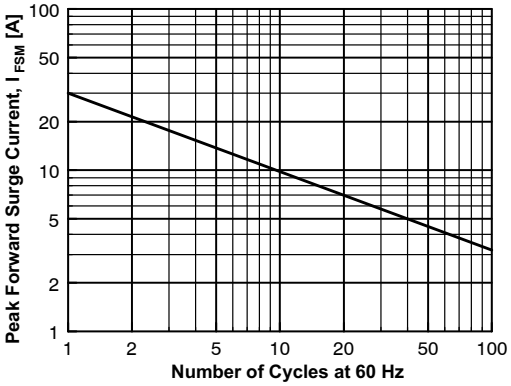


Figure 2. Non−Repetitive Surge Current

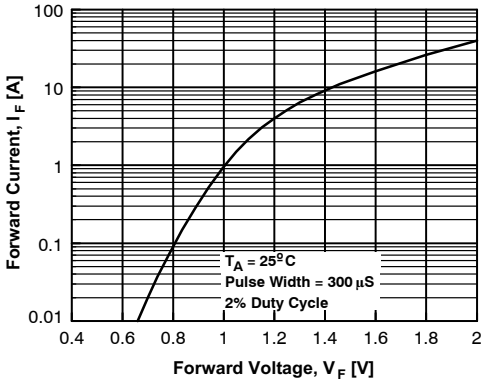


Figure 3. Forward Voltage Characteristics

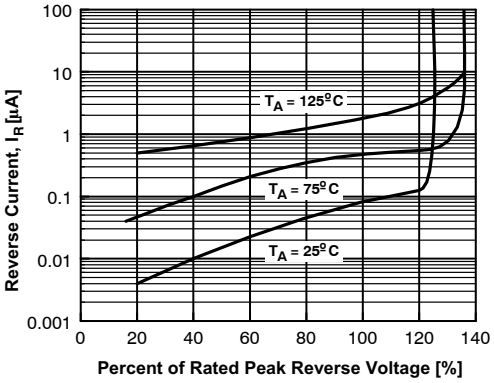


Figure 4. Reverse Current vs. Reverse Voltage

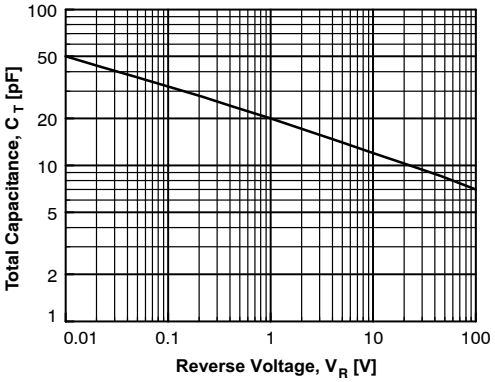
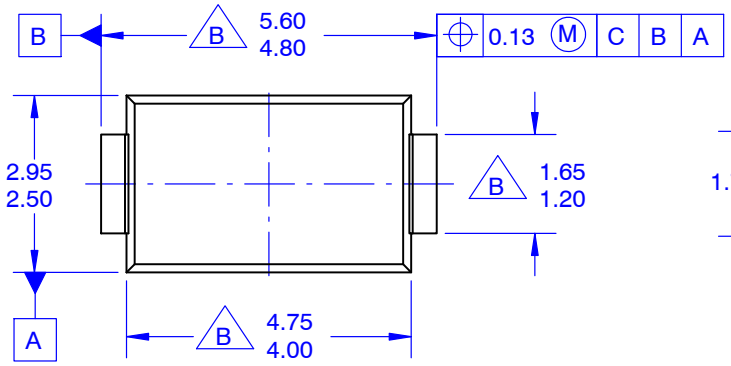


Figure 5. Total Capacitance

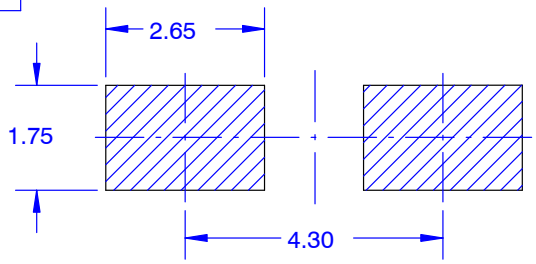
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ISSUE O

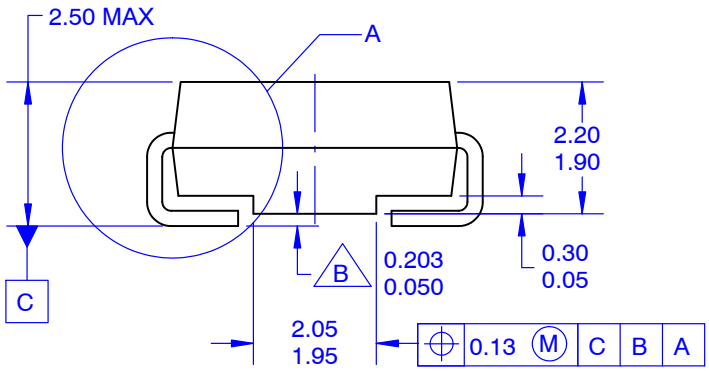
DATE 31 AUG 2016



TOP VIEW



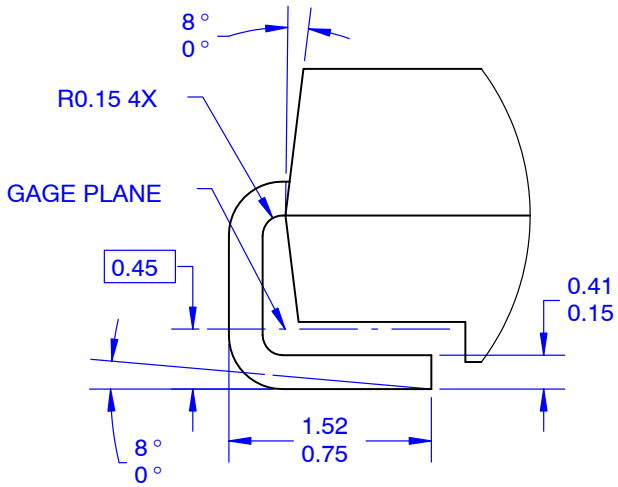
LAND PATTERN RECOMMENDATION



SIDE VIEW

NOTES:

A. EXCEPT WHERE NOTED, CONFORMS TO JEDEC DO214 VARIATION AC.
B DOES NOT COMPLY JEDEC STANDARD VALUE.
C. ALL DIMENSIONS ARE IN MILLIMETERS.
D. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR PROTRUSIONS.
E. DIMENSIONS AND TOLERANCE AS PER ASME Y14.5–2009.
E. LAND PATTERN STD. DIOM5025X231M



DETAIL A
SCALE 20 : 1

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